

Abhishek Janagoudar

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📄 Visa Valid till Feb 2027



Profile

Master's student in Electrical and Information Technology, specializing in Automation and Robotics at Hochschule Darmstadt. Gained hands-on experience in mobile robotics during a full-time mandatory internship at Baer Automation. Actively seeking working student, internship, and thesis opportunities.

Professional Experience

BÄR Automation GmbH, Mobile Robots Development Intern

Full-time Mandatory Internship

11/2025 – Present
Stuttgart, Germany

- **Evaluated and implemented SLAM algorithms:** Configured SLAM Toolbox and Google Cartographer in ROS 2, conducting comparative analyses across multiple mapping modes (e.g., async, lifelong) to optimize environmental mapping for mobile robots.
- **Optimized robot localization:** Fine-tuned sensor configurations, adjusting LiDAR scan matching and odometry parameters to significantly increase localization accuracy and mitigate drift over time.
- **Worked with AGV VDA 5050 standards.**
- **Built digital twin tools:** I had the opportunity to learn about Asset Administration Shells (AAS v3.1). I helped create a tool that automatically generates these digital assets and submodels for the ARENA2036 project using official AAS3.1 standards and IDTA Templates for Submodels, making sure all the required technical attributes were included correctly.

J. Ray McDermott Engineering Services Private Ltd

Associate System Engineer - AVEVA Electrical Administrator V12

02/2024 – 08/2024
Chennai, India

- Administered AVEVA Electrical V12 for Ruya EPCIC offshore projects 11 & 13, delivering 50+ electrical datasheets and drawings on time.
- Standardized project workflows that reduced rework hours and improved inter-team coordination.
- Led tool training sessions, reducing internal troubleshooting incidents by 30% within 5 months.

Infosys Ltd.

Senior System Engineer - AVEVA Engineering V15 Electrical & Instrumentation Administrator

09/2021 – 02/2024
Mysore, India

- Managed and optimized AVEVA Engineering 15.x for Electrical and Instrumentation projects, including application architecture, operations, and system maintenance.
- Acted as Technical Scrum master, coordinating sprint reviews, retrospectives, and daily stand-ups, promoting Agile practices.

System Engineer:

- Configured libraries and settings, including TAS, I/O cards, wiring rules, instruments, and cable catalogs.
- 50+ User Story deliverables (Electrical Load list, Cable Schedule, Reports, Datasheets, and Drawings).
- Developed data model with classes, attributes, associations, LOVs database views, and Kraftwerk Kennzeichen System naming conventions.
- Mentored 9 engineers through recurring technical and application training sessions and supported project builds, deployments, and application setups.

Arkay Energy Ltd.

Electrical Operations and Maintenance Intern

Supported commissioning/testing of substation systems, performed Operations and Maintenance tasks, learned industrial safety.

07/2019 – 07/2019
Ramanathapuram,
India

LARSEN AND TOUBRO ELECTRICALS

Electrical Commissioning Intern

Assisted in electrical system setup/testing, troubleshooting, and system optimization.

06/2019 – 06/2019
Khargone, India

Education

Master of Science (M.Sc) | Electrical and Information Technology,
Darmstadt University of Applied Sciences

09/2024 – present
Darmstadt,
Germany

Course Overview:

Industrial Robotics, Safety In Industrial Automation, Computer Vision, Industry 4.0 Industrial Internet of Things (IIoT) Adaptive Control, Modelling and Identification, Advanced Programming Techniques, State Space Analysis, Human Machine Interface, Advanced Sensors

Bachelor of Engineering (B.E) | Electrical and Electronics, *The National Institute of Engineering*
Grade: 8.12 CGPA (**German Grade:** 1.7)

07/2017 – 09/2021
Mysore, India

Projects

Robotics and Computer Vision Projects

- 3-DOF Robotic Arm: Built and programmed a 3-DOF robotic arm with servo joints, implementing forward and inverse kinematics, and created a Python interface for Arduino-PC communication.
- Point Cloud Segmentation: Developed a pipeline in Python and ROS2 for indoor mapping using a Hesai XT16 LiDAR on a UniTree quadruped robot, applying basic statistical and machine learning methods for feature extraction.
- Visual Servoing (Igus ReBeL 6DOF): Used OpenCV template-matching to map image errors to robot joint commands for closed-loop visual control.
- UR5/UR10 Kinematics: Built MATLAB/Simulink models to calculate forward and inverse kinematics and analyze manipulator motion under external loads.

Advanced Programming Technique Lab Projects

- BattleShip Game: Developed a console-based Battleship game in C++ to enhance programming skills, focusing on game logic and user interaction.
- Student Database: Developed and designed an optimized database system for university enrollment using C++ (object-oriented programming), ensuring efficient data management and retrieval.
- Reversi Board Game: Implemented a console-based Reversi board game in C++ to strengthen algorithmic thinking and data handling.

Safety in Industrial Automation Lab:

- Programmed Siemens S7-1516F safety PLC systems and ET200SP modules using TIA Portal, implementing critical safety logic and functions such as ESTOP1, TWO_H_EN, SFDOOR, and FDBACK.
- Conducted rigorous discrepancy analyses and black box testing to ensure safety integrity.
- Evaluated safety functions according to EN ISO 13849-1 standards using SISTEMA software, calculating PerformanceLevels (PLr), verifying compliance, and enhancing system reliability and safety.

Languages

English

IELTS - 7.5 (C1)

Deutsch

A2.2

Hindi

Native

Kannada

Native

SKILLS

Robotics & Mobile Robots: Robot kinematics, Gazebo, RViz, SLAM (SLAMToolbox, Cartographer, AMCL), ROS2, AGV navigation, VDA 5050 compliance, Linux, Git, Docker.

Computer Vision: Image Preprocessing, Camera Calibration, Object Detection, Segmentation, Multi-Object Tracking, LiDAR & Point Cloud Processing

Industrial Internet of Things (IIoT): OPC UA, Node-RED, MQTT, IoT Dashboards, IoT Gateway Integration, Digital Twins, Virtual Commissioning

Automation: TIA Portal, Siemens S7 Safety, PLC Programming, SISTEMA

Programming: Python, C++

Soft Skills: Project Management, Agile/Scrum, Team Collaboration, Technical Training, Troubleshooting, Teamwork, CI/CD pipeline

Other: Safety in the Utility Industry, Operations, Maintenance, and Commissioning

Engineering Tools: AVEVA Engineering v15.x, AVEVA Electrical v12, Jira, Rally, Software testing